

Derrick H. Sun

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Summary: M.S. in Computer Science with software engineering skills, industry internship, and research in machine learning. Experience automating workflows, building predictive models, and optimizing AI architectures. US Citizen. Available to work immediately, open to relocation.

Education

- ❖ **Purdue University, M.S. in Computer Science** **August 2024 – May 2026**
 - Specialization in AI and Machine Learning with focus on Computer Security
 - Advanced courses: Deep learning, Prompt Engineering (DSPy), and LLM integration
- ❖ **UC Berkeley, B.A. in Computer Science and in Applied Math** **June 2020 – May 2024**
 - **Double major** with emphasis on machine learning, AI, and statistics
 - Courses: Deep Learning, Statistical Analysis, Data Structures & Algorithms

Work Experience

- ❖ **Teaching Assistant | [Purdue University](#)**, West Lafayette, IN. **August 2024 - May 2025**
 - Led discussion sessions and held office hours for undergraduate CS courses.
- ❖ **Data Scientist Intern | [Bridger Investment Partners](#)**, New York, NY. **May 2024 – August 2024**
 - Automated data processing workflows with Prefect, removing manual workflows.
 - Ran statistical analysis of mortgage data with Python & SQL on AWS S3 & Redshift.
- ❖ **Research Assistant | [UC Berkeley](#)**, Berkeley, CA. **December 2023 – May 2024**
 - Conducted comparative analysis of transformer architectures (GPT-2, Mamba, LLaMA) on in-context learning tasks.
 - Results available on [GitHub](#) ([code](#) and [arXiv](#)).
- ❖ **iOS Developer Intern | [NewsBreak](#)**, Seattle, WA. **May 2022 – August 2022**
 - Implemented the Discover page of the NewsBreak iOS app using Swift and UIKit.
 - Developed data-fetching to retrieve & display news for user-selected organizations.

Research and Publications

- ❖ Research Coauthor | **UC Berkeley**, [ICL with Custom Models](#) **December 2023 – May 2024**
 - Conducted research on hybrid LLM architectures and evaluated their performance
- ❖ Research Coauthor | **UCSD**, [Skewed Random Utility Models](#) **May 2023 – May 2024**
 - Employed Gosper's hack and memoization to improve computational speed by 80%.

Projects

- ❖ **[Quickboard Firefox Extension](#)** **June 2026**
 - Developed and published a Firefox extension providing a persistent toolbar for storing and quickly pasting reusable text snippets.
- ❖ **[Job Search Assistant](#)** **May 2026 - Present**
 - Built an automated job-search pipeline that filters listings based on experience requirements, company characteristics, and other configurable user criteria
 - Automated completion of routine application fields, saving substantive questions for manual review; developed a Firefox extension for interactive use
 - Integrated AWS S3 for application tracking, with multi-device compatibility
- ❖ **[Webnovel Downloader and Translator](#)** **2025 – 2026**
 - Personal project that downloads online content and calls OpenAI API to translate it.
 - Utilizes Selenium to login, display a menu and automatically navigate the internet.
- ❖ **[Transformer Optimization \(Report\)](#)** **Fall 2023**
 - Tested transformer performance on various simple tasks with ReLU-attention and quantization optimizations, reducing model runtime on simple tasks by 0.1 seconds.

Programming Skills

- ❖ Programming Languages: Python, Java, SQL, C, C++, R, Swift, JavaScript, TypeScript HTML/CSS
- ❖ ML/AI Frameworks: PyTorch, TensorFlow, DSPy, Hugging Face
- ❖ Data Tools: Pandas, NumPy, Matplotlib, Seaborn, RStudio, Jupyter, Scikit-learn
- ❖ Cloud & Databases: AWS S3, Redshift, PostgreSQL, MongoDB
- ❖ Development Tools: Git, Docker, Prefect, XCode, LaTeX, Bootstrap, Selenium, Node.js